Muhammad Umer Bappi

Computer Science Graduate

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Professional Summary

Computer Science graduate with hands-on experience across the full software development lifecycle, from system design and implementation to testing, deployment, and production. Specialises in Machine Learning and NLP, with a proven track record of building end-to-end AI systems that deliver measurable results. Brings strong applied engineering instincts, a research-driven approach to problem solving, and the ability to pick up new technologies and contribute quickly across diverse technical environments. Equally comfortable working on research-oriented challenges and product-focused engineering.

Education

Bachelor of Science in Computer Science
Forman Christian College (University), Pakistan — December 2025
Key Courses:
Data Structures & Algorithms, Operating Systems, Databases, Networks, AI, Machine Learning, NLP, Game Development, Information Security, Discrete Mathematics, Java.

Professional Experience

Revotic AI – Remote
ML/AI Intern | September 2025 – November 2025

- Designed and built a machine learning-based weather forecast system, owning the full pipeline from data preprocessing to model evaluation and delivery
- Contributed to end-to-end development of production AI solutions, collaborating with engineers on real-world applied ML problems
- Gained hands-on experience translating business requirements into technical ML specifications within a fast-paced startup environment

Forman Christian College (University) – Lahore, Pakistan
Research Assistant | February 2024 – December 2025

- Identified research gaps in the ML/NLP domain and independently developed novel research directions, demonstrating initiative and scientific rigour
- Designed and executed experimental frameworks to validate hypotheses, applying cutting-edge methodologies from current literature
- Collaborated with faculty supervisors on manuscript preparation, contributing to academic publications and advancing research output

Technical Skills

Languages & Frameworks: Java, Python, C++, React, Node.js
Databases: PostgreSQL, MySQL
DevOps & Cloud: Docker, AWS, GCP, CI/CD
Tools & Methodologies: Git, Confluence, Jira, REST APIs, Clean Code
ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, PEFT, Unsloth, MLX
LLM & NLP: LLM Fine-Tuning (LoRA, QLoRA), Prompt Engineering, RAG, Agentic Architectures, Text Generation, Langchain, FAISS
Evaluation & Data: BERTScore, Exact Match, Data Augmentation, Multi-format Processing (JSON/JSONL/CSV), Pandas, NumPy
Other: WebRTC, FaaS (AWS Lambda, Apache OpenWhisk), Unit & Performance Testing, Liquibase

Projects

- **LLMs & Factuality in Urdu NLP — Final Year Project**
  Built a novel multi-agent LLM framework achieving a 35% reduction in hallucinations in Urdu NLP, using PEFT fine-tuning, RAG, and Agentic architectures; research paper submitted to international conferences.
- **LyricGen — AI-Powered Lyric Generation from Audio Analysis**
  Built an end-to-end audio-to-lyrics generation system by fine-tuning Llama 3.2-3B via LoRA on a curated Million Song Dataset subset, extracting audio features using Librosa and deploying an interactive Gradio web application.
- **Real-Time Face Detection & Age Estimation System**
  Built a real-time face detection and age estimation system using OpenCV and DeepFace, with a Random Forest model for age prediction and live video stream integration.
- **Custom Tower Defence Game — Godot**
  Developed an original tower defence game in Godot featuring a custom story plot, original scenes, and full gameplay mechanics from scratch.
- **FocusScaffold — AI-Powered ADHD Productivity Web Application**

Built a full-stack executive-function support tool with AI-driven task decomposition, cognitive dump workflows, focus session tracking, and a weekly pattern dashboard, designed to reduce cognitive load for users with attention and task-initiation challenges.

Awards & Honors

- **Best Poster Award – COMSPEX Forman Christian College (University) - 2025**
  Recognised for research presentation on LLM hallucination in Urdu NLP.
- **Music Coordinator, Forman Music Society (FMS) 2022 - 2024**
  Led event coordination and production management for university-wide musical programmes.
- **Battle of Bands Winner – CMH Lahore BoB 2024**
  1st Place, Battle of the Bands inter-university competition.

Extracurricular Activities

- Music: Session Lead Guitarist, Original Music, Music Producer
- Sports: Football, Swimming, Table Tennis
- Reading: Technical books and blogs